

1 Agroforestry SmartAgroforst: Poplar biomass modelling

2 Thomas Kirschstein^{1,1,1,*}

3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

6 1. Introduction

7 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
8 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
9 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
10 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such

*Corresponding author

Email address: thomas.kirschstein@ikts.fraunhofer.de (Thomas Kirschstein)

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11 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
12 proving soil quality and providing multiple ecosystem services compared to conventional
13 monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sus-
14 tainable feedstock and can be used either energetically, chemically or materially (e.g. in
15 the building sector). The latter two options are generally preferred over the energetic
16 usage from a cleaner production perspective as biogenic carbon is stored in products
17 during their life span which contributes to climate change mitigation.

18 Despite substantial research on ecological advantages and pioneering initiatives to
19 establish AFS in practice, the share of AFS in agricultural production systems is still
20 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
21 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
22 challenges in establishing AFS systems at large scale in practice is a lack of opportunities
23 to generate substantial revenues from woody biomasses of AFS. On the other side, the
24 rise of bioeconomy creates substantial demands for woody biomass and requires a reliable
25 and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low
26 energy density, the design of cost-efficient supply chains is crucial to ensure the economic
27 viability of AFS-based value chains. This requires coordinated decisions on stand estab-
28 lishment, multi-cycle harvesting, and product cascading in spatially distributed supply
29 chains, which can be supported by integrated optimisation models that link stand-level
30 growth dynamics to regional supply chain design.

31 In the last two decades, a large body of research has analysed biomass supply chains
32 for bioenergy and biorefineries using mathematical programming models, often in the
33 form of mixed-integer linear programming (MILP) (Zahraee et al., 2019; Sharma et al.,
34 2013). These models typically optimise network configuration, feedstock sourcing, and
35 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
36 nous input or rely on simplified agricultural production decisions. Moreover, downstream
37 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
38 consuming all produced biomasses. Recent work has started to integrate operational de-
39 tails such as various biomass types, biomass growth models and regeneration processes
40 in multi-period supply chain models.

41 The present paper contributes to this literature in four ways. First, we propose

42 a novel MILP formulation for AFS-based biomass supply chain design that explicitly
43 links establishment decisions and consecutive harvest decisions, ensuring that harvesting
44 decisions at each site are consistent with minimum and maximum rotation ages. Second,
45 we embed allometric yield functions for different biomass types generating age-specific
46 yields for different woody biomass types (like stems, branches, or leaves and twigs). Third,
47 these different woody biomass types can be processed in specific downstream industries,
48 like chemical or paper industry as well as energy sector. The model aims to provide
49 strategic decision support for establishing regional AFS-based bioeconomy ecosystems
50 connecting multiple industries and product markets.

51 The remainder of the paper is structured as follows. Section 2 reviews the relevant
52 literature on agroforestry biomass systems, allometric biomass modelling, and biomass
53 supply chain optimisation. Section ?? introduces the allometric models adopted to rep-
54 resent biomass growth of different tree species and biomass types. Section 3 summarises
55 the MILP formulation for the agroforestry supply chain with age-lag constraints. Sec-
56 tion 4 outlines the planned computational experiments for the case study and discusses
57 expected insights for cleaner production strategies.

58 **2. Literature review**

59 Agroforestry biomass systems, stand-level growth models, and biomass supply chain
60 optimisation have each been studied extensively, but they are rarely combined in an inte-
61 grated framework that links age-dependent stand dynamics to regional product-cascading
62 value chains (Zahraee et al., 2019; Sharma et al., 2013; Espinoza et al., 2017). This sec-
63 tion reviews (i) agroforestry systems with emphasis on system types, economics, and the
64 cascading use of biomass as the primary revenue driver; (ii) biomass growth models at
65 tree and stand level with explicit attention to the allometric–diameter–stand–biomass
66 pathway; and (iii) biomass supply chain design using mathematical programming. The
67 section concludes by identifying the modelling gap addressed in this paper.

68 *2.1. Agroforestry systems*

69 Agroforestry deliberately integrates woody perennials with crops and/or livestock on
70 the same management unit, with the aim of enhancing productivity, profitability, and

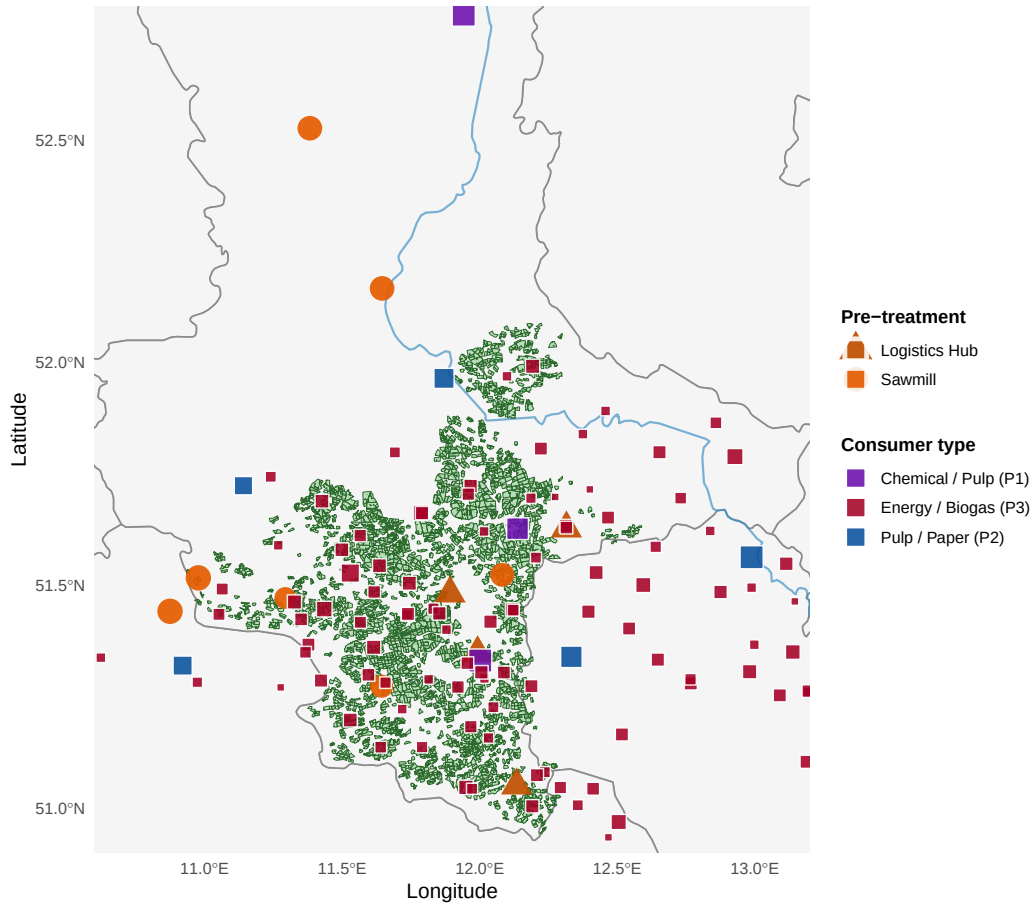


Figure 1: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Green polygons: potential KUP sites. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

71 environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three
72 distinct system types can be distinguished based on the spatial arrangement of trees and
73 the rotation length at which woody biomass is harvested.

74 **Short-rotation coppice (SRC)** is defined as a monoculture plantation of fast-
75 growing woody species covering the entire management unit, with planting densities
76 of 8,000–20,000 cuttings $\cdot\text{ha}^{-1}$. After each harvest the root system regenerates without
77 replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years
78 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
79 but it is helpful to benchmark biomass production.

80 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
81 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise
82 arable field in an alley-cropping arrangement. Crop production continues in the inter-
83 row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
84 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are
85 comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light
86 and water availability than interior SRC trees. This boundary effect raises individual-tree
87 biomass accumulation but also intensifies competition with adjacent crop rows (Huber
88 et al., 2018).

89 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spa-
90 tial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al.,
91 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem di-
92 ameters and the proportion of high-value timber increases markedly, enabling premium
93 material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-
94 rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry,
95 oak, or high-grade poplar clones) are planted.

96 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age
97 is hard to define and, more importantly, the intended use of AFS biomass is not always
98 clear and can also change over time. Nonetheless, when establishing AFS, the structure
99 of AFS and its intended use after harvest drives planting decisions and, subsequently,
100 economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as
101 separate AFS systems, but the interplay of intended use, establishment, and harvesting

102 decisions is analysed.

103 2.1.1. *Economic rationale of landowners*

104 The decision of a landowner to establish an AFS involves weighing substantial up-front
105 costs and long-term opportunity costs against a diversified revenue stream. On the **cost**
106 **side**, establishment requires site preparation, planting material, and planting labour,
107 typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude,
108 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping,
109 transport to roadside) add approximately 200–400 €/ha per rotation depending on stand
110 density and equipment availability (Testa et al., 2014; Spinelli et al., 2011). During
111 the rotation, maintenance cost of AFS arise on a yearly basis. Next to expenses for
112 establishment, maintenance and harvesting, opportunity cost arise as no revenues from
113 crops are generate on tree strips. In regions with fertile soils and high cereal prices, yearly
114 crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves et al.,
115 2007). Thus, opportunity cost are often the dominant cost driver in SRAFS profitability
116 analyses .

117 On the **benefit side**, the primary revenue of AFS is the sale of harvested woody
118 biomass, whose magnitude depends critically on rotation length, species, and site pro-
119 ductivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25
120 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk
121 et al., 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS
122 and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotran-
123 spiration; they provide partial shading that moderates heat stress during drought years;
124 and their root systems may improve water retention in the topsoil (Graves et al., 2007;
125 Grünewald et al., 2007). Empirical measurements in Central European alley-cropping
126 trials indicate that these microclimate effects can increase cereal yields in adjacent strips
127 by 5–15% relative to open-field controls, partially or fully compensating the loss of arable
128 area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS
129 deliver broader ecosystem services (like soil organic matter accumulation, carbon se-
130 questration, biodiversity enhancement) that may lead to agri-environment subsidies and
131 carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

132 Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS

133 biomass. Thereby, revenues are different for the types of biomass and in which value
 134 chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power
 135 generation, or anaerobic digestion, which typically creates low prices per tonne of biomass.
 136 Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also
 137 be used as pulp and paper grade material as well as feedstock for chemical industry
 138 or as timber achieving higher prices (Lamers et al., 2015; De Meyer et al., 2015). Thus,
 139 fostering AFS establishment by maximizing economic returns from AFS biomass requires
 140 a strategic approach to product cascading that allocates different biomass fractions to
 141 the highest-value applications, while ensuring that the rotation length is aligned with
 142 the growth dynamics of the stand and the market demand for different biomass types
 143 (Lamers et al., 2015).

144 Thus, for making educated decisions on AFS establishment and harvesting, the ex-
 145 pected biomass quantities at harvest for stem, branches, and residues need to be esti-
 146 mated. Thereby, the shares of the biomass types change with stand age. E.g.~the share
 147 of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018;
 148 Testa et al., 2014). Therefore, in Section 2.2 growth modelling approaches are reviewed
 149 that capture the age-dependent growth dynamics of different biomass types in AFS.

150 2.2. Biomass growth models

151 Total biomass on tree level is commonly modeled using allometric functions that
 152 relate tree characteristics like stem diameter or height to total biomass weight. There are
 153 numerous allometric models in the literature, often specific to species, site conditions, and
 154 stand age. For fast-growing species used in SRAFS (like poplar and willow), allometric
 155 functions of the power-law form are used [Reference]. I.e., dry biomass M_i (kg) of a shoot
 156 i is approximated by

$$M_i = b_0 \cdot SBD_i^{b_1} \quad (1)$$

157 where SBD is the shoot's stem base diameter (in cm) measured 10 cm above ground.
 158 Parameters b_0 and b_1 are species-specific (Huber et al., 2016). In the following it is
 159 assumed that each tree is pruned to one dominant shoot. For SRAFS species in Southern
 160 Germany, Huber et al. (2018) reports that most trees produce not more than 2 shoots
 161 and Huber et al. (2016) report a common exponent $b_1 = 2.603$ for black alder, black

locust, poplar (Max~3), and willow (Inger), while b_0 ranges from 0.025 to 0.041, reflecting differences in wood density and branching structure.

Allometric rules like (1) generate estimates of biomass quantities for a given shoot, i.e. they can be used to support operational harvest decisions. For planning on a strategic level, it is necessary to estimate biomass development over time, i.e. an SBD growth model. The mean stem base diameter $\overline{\text{SBD}}(t)$ of coppice shoots increases rapidly in the first growing seasons as shoots exploit stored root carbohydrates, then decelerates as intra-stand competition for light and water intensifies. Such trajectories can be described by sigmoidal growth functions. Empirical evidence from poplar SRC and SRAFS trials indicates that the SBD growth curve exhibits maximal growth rate (i.e., inflection point) at a relatively early stand age (2–4 years) (Barrio-Anta et al., 2008; Niemczyk et al., 2021). Therefore, a Gompertz model for $\overline{\text{SBD}}(t)$ can be formulated as

$$\overline{\text{SBD}}(t) = D_\infty \cdot e^{-e^{-r \cdot (t-\tau)}}, \quad (2)$$

where D_∞ (cm) is the asymptotic mean diameter, r (yr^{-1}) controls the rate of diameter increment, and τ (yr) is the age at the inflection point of the diameter curve. Given $\overline{\text{SBD}}(t)$ from (2), the mean shoot biomass growth follows directly from the allometric equation (1):

$$\bar{M}_{\text{shoot}}(t) = b_0 \cdot \overline{\text{SBD}}(t)^{b_1} = d_\infty \cdot b_0 \cdot e^{-e^{-r \cdot b_1 \cdot (t-\tau)}}. \quad (3)$$

Thereby, $b_\infty = d_\infty \cdot b_0$ denotes the asymptotic mean shoot biomass, and the effective growth rate of shoot biomass is $b'_1 = r \cdot b_1$, which is amplified relative to the biomass growth rate r by the allometric exponent b_1 .

Total aboveground biomass quantities on stand level $M_{\text{AG}}(t)$ (in t per ha) is then obtained by aggregating individual shoot biomasses over all trees and multiplying by stand density $N(t)$ (in trees per ha):

$$M_{\text{AG}}(t) = N(t) \cdot \bar{n}_{\text{shoot}}(t) \cdot \bar{M}_{\text{shoot}}(t), \quad (4)$$

where $\bar{n}_{\text{shoot}}$ is the mean number of shoots per tree (Huber et al., 2018). Both $\bar{n}_{\text{shoot}}(t)$ and $N(t)$ may change over time. The number of shoots is often controlled in tree care processes and reduced to 1 or 2 shoots per tree (Huber et al., 2018). Stand level density $N(t)$ is initially defined when the AFS is established and often declines somewhat over

time, but usually saturates quickly [REFERENCE]. For the product $M_{AG}(t)$ in (4) a right-skewed sigmoidal shape with an inflection occurring at a relatively low fraction of the asymptote can be assumed which can be captured in a Gompertz model as well (Niemczyk et al., 2021; Testa et al., 2014). Assuming constant tree density and an average shoot number of 1, $M_{AG}(t)$ can be simplified as

$$\bar{M}_{AG}(t) = a_{\infty} \cdot e^{-e^{-b'_1 \cdot (t-\tau)}}. \quad (5)$$

where $a_{\infty} = N \cdot \bar{n}_{shoot} \cdot d_{\infty} \cdot b_0$ is the asymptotic stand-level biomass. For poplar clone Max~3 under SRAFS conditions in Central Europe, representative parameter values are $D_{\infty} = 9.5$ cm, $r = 0.55$, and $\tau = 2.5$ years, yielding a mean SBD of approximately 4–5 cm at age 3 and 7–9 cm at age 7 which is consistent with the diameter ranges reported by Huber et al. (2018) and Barrio-Anta et al. (2008). For total aboveground biomass of poplar clone Max 3 under SRAFS management, the Gompertz model (5) is parameterised as $a_{\infty} = 100$ t of dry biomass per ha, $b'_1 = 0.32$ per year, and $\tau = 4.0$ years, yielding a growth peak at approximately age 4 and a total aboveground biomass of roughly 65 t of dry biomass per ha at age 10 which is consistent with the empirical reference range of 50–135 t per ha reported for comparable poplar systems in the literature (Huber et al., 2018; Niemczyk et al., 2021).

Yet, for the purpose of modelling AFS supply chains, the three biomass components merchantable stem, structural branches, and fine residues need to be distinguished. Thus, the total biomass growth model (5) needs to be adjusted those types. Thereby, growth characteristics are different each type and, thus, require parameterization. Thereby, the differentiation between the merchantable stem and branch depends on the requirements defined by downstream processes. E.g. for chemical production, the merchantable stem is defined as the portion of the tree with a certain minimum diameter (e.g. 10 cm), while branches and residues are defined as the portion of the tree that cannot be chipped into such large chips and thus require different processing or are used for lower-value applications. The share of these components in total aboveground biomass changes with stand age, which has important implications for product cascading decisions in AFS-based supply chains.

Empirical studies show that the stem fraction rises from roughly 55% of AGB at age~4 to over 75% at age~10, while branch and residue shares decline as stands mature

(Huber et al., 2018; Testa et al., 2014, ?). This age-dependence is biologically driven by the increasing dominance of secondary stem growth (radial increment) relative to primary shoot elongation and branching in later stand development stages. To capture these dynamics analytically, each component share $f_p(t)$ is represented as a normalised Gompertz function:

$$h_p(t) = a_p \cdot e^{-e^{-b_p \cdot (t - \tau_{0,p})}} \quad (6)$$

$$f_p(t) = \frac{h_p(t)}{\sum_{p'} h_{p'}(t)}. \quad (7)$$

For Max~3 the stem helper is parameterised with a high asymptote ($a_{\text{stem}} = 1.0$) and a relatively late inflection ($\tau_{0,\text{stem}} = 4.0$ years), whereas the branch uses a lower asymptote ($a_{\text{branch}} = 0.7$) and an earlier inflection ($\tau_{0,\text{branch}} = 3.0$ years), and the residue helper saturates earliest ($a_{\text{rest}} = 0.1$, $\tau_{0,\text{rest}} = 2.0$ years). This parameterisation reproduces the observed shift towards stem dominance with age and is consistent with the empirical component fractions compiled across poplar agroforestry and SRC systems spanning ages 6–13 years, as illustrated in Figure 2.

2.3. Biomass supply chain design

Biomass supply chain design has been extensively studied using MILP at strategic, tactical, and operational levels (Zahraee et al., 2019; Sharma et al., 2013; Espinoza et al., 2017). Typical models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Eksioglu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infrastructure and operational decisions across seasonal or annual time steps (Eksioglu et al., 2009; Zahraee et al., 2019).

Among generic multi-product frameworks, OPTIMASS explicitly models product quality cascades by defining a hierarchy of biomass classes and allowing higher-quality products to satisfy lower-quality demands, while the t-OPTIMASS extension incorporates biomass growth and regeneration over multiple periods (De Meyer et al., 2015, 2016). However, in virtually all existing formulations biomass availability is treated as

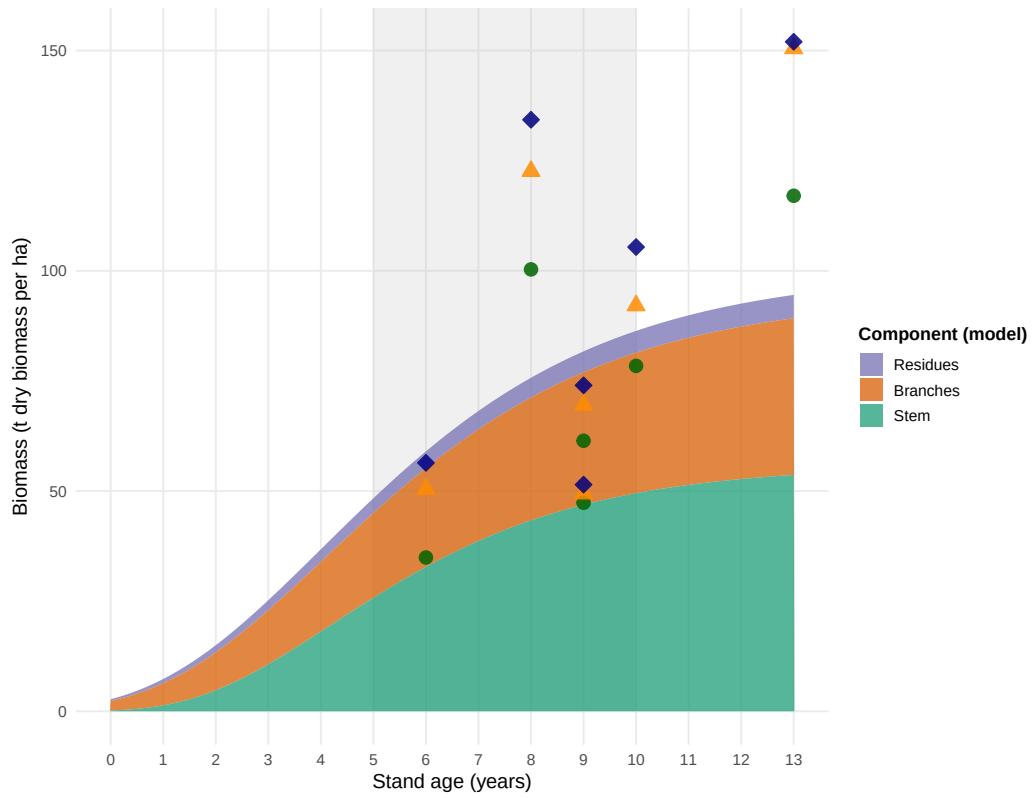


Figure 2: Modelled stand-level biomass components for poplar (Max 3) over stand age, represented as stacked Gompertz curves (shaded areas: stem green, branches orange, residues purple) and compared with empirical aboveground biomass observations from poplar agroforestry and SRC systems. Symbols denote cumulative component totals: stem (filled circles), stem+branches (triangles), total AGB (diamonds). Empirical data compiled from Jha (2018) and cited primary sources. Grey shading: typical rotation-age window (5–10 years).

an exogenous time-series input: stand age structure, rotation decisions, and coppice cycles are not represented endogenously, so that trade-offs between rotation length, stand management, and supply chain configuration cannot be explored in a unified model (De Meyer et al., 2016; Lamers et al., 2015). This gap is particularly pronounced for agroforestry systems with repeated coppice cycles, where establishment timing, consecutive harvest decisions, and age-dependent biomass quality jointly determine both the feasible supply profile and the achievable product cascade.

The MILP formulation developed in this paper addresses this gap by endogenising stand age dynamics through binary age-lag variables, coupling the component-specific yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain flow variables, and embedding a product quality cascade that allocates stem, branch, and residue fractions to chemical, pulp, and energy markets within a single optimisation model.

3. MILP model for agroforestry supply chain design

3.1. Sets, indices, and time structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To represent stand establishment and harvest timing, extended time sets $\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$ are defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest. The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \dots, T_{\max} + 1\}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}. \quad (8)$$

3.2. Decision variables

Binary variables z_{ist} indicate whether, at site i , periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$ as defined in (8). These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t)

of product p at site i in harvest period t . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

3.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned}
\max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \\
& - \sum_i C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} z_{i0t} \\
& - \sum_i C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\
& - \sum_{i,(s,t) \in \mathcal{S}: s > 0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\
& - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}.
\end{aligned} \tag{9}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare (Testa et al., 2014). Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

3.4. Key constraints

The path connectivity constraint ensures that each harvest period at a site has exactly one predecessor and one successor in the age-lag graph, effectively forming a path over $\mathcal{T}^+$:

$$\sum_{s=0:(s,t) \in \mathcal{S}} z_{ist} = \sum_{u=t+1:(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \tag{10}$$

Establishment is restricted to at most one time per site:

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I}. \tag{11}$$

288 Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\text{harv}}. \quad (12)$$

289 Harvest quantities bound outbound flows from each site:

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}. \quad (13)$$

290 At storage hubs, inventories obey standard balance equations with initial inventory
291 zero:

$$S_{jpt} = S_{jp,t-1} + \sum_i X_{ijpt} - \sum_{k,p' \geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\text{harv}}. \quad (14)$$

292 Storage and processing capacities are enforced via:

$$\sum_p S_{jpt} \leq \text{CAP}_j^{\text{stor}} \quad \forall j, t, \quad \sum_{i,p} X_{ijpt} \leq \text{CAP}_j^{\text{proc}} \quad \forall j, t. \quad (15)$$

293 Demand satisfaction with cascades is modelled as described above, where higher-
294 quality products can satisfy lower-quality demands up to maximum demand levels D_{kpt}^{max} .

295 4. Case study and computational experiment plan

296 The proposed MILP model is intended to support the design of agroforestry biomass
297 supply chains for cleaner production in specific regional contexts. In a forthcoming case
298 study, we focus on poplar-based alley-cropping systems in a temperate European setting,
299 informed by empirical data from SRAFS trials in Southern Germany and SRC plantations
300 in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

301 We envisage the following steps and experimental scenarios for computational analy-
302 sis:

- 303 1. **Parameterisation of allometric yield profiles:** Use species- and site-specific
304 allometric models and observed growth trajectories to construct age-dependent
305 yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative
306 agroforestry sites. Where necessary, scenario ranges will be defined to capture
307 uncertainty in MAI and CAI due to climate variability and management differences
308 (Huber et al., 2016, 2018; Trnka et al., 2008).

- 309 2. **Baseline optimisation:** Solve the MILP model (9)–(15) for a baseline configu-
310 ration with fixed prices, capacities, and policy settings, obtaining optimal estab-
311 lishment timing, harvest schedules, and supply chain structure (locations, flows,
312 inventories).
- 313 3. **Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ in (8) to
314 represent alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse
315 their impacts on profitability, product mix, and land use. This will shed light on
316 trade-offs between shorter rotations favouring energy chips and longer rotations
317 enabling higher shares of material products (Trnka et al., 2008; Testa et al., 2014).
- 318 4. **Product price and cascading scenarios:** Explore scenarios with different rel-
319 ative prices for chemical, pulp, and energy products, reflecting changes in market
320 demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets).
321 Assess how the product cascade hierarchy and yield quality structure influence the
322 marginal value of species and rotation decisions (De Meyer et al., 2015; Lamers
323 et al., 2015).
- 324 5. **Infrastructure and logistics scenarios:** Evaluate the effect of alternative stor-
325 age and processing network configurations (e.g., decentralised vs. centralised hubs,
326 different capacity levels, transport cost parameters) on the optimal spatial deploy-
327 ment of agroforestry sites and biomass flows (De Meyer et al., 2015, 2016; Lamers
328 et al., 2015).
- 329 6. **Sensitivity to allometric and growth uncertainty:** Conduct parametric sen-
330 sitivity and, where computationally feasible, stochastic analyses to quantify the
331 impact of uncertainty in allometric parameters and growth responses (e.g., under
332 drought or management differences) on optimal decisions and system robustness
333 (Huber et al., 2018; Trnka et al., 2008).
- 334 7. **Cleaner production indicators:** For selected solutions, compute derived indica-
335 tors such as biomass cascading index (share of biomass used for material products
336 before energy), utilisation rates of higher-quality fractions, and, where coupled with
337 LCA data, greenhouse gas emission savings relative to fossil benchmarks (Espinoza
338 et al., 2017; Zahraee et al., 2019).

5. Conclusions

This paper outlines a research framework that couples allometric biomass growth models (1)–(7) with a MILP-based agroforestry biomass supply chain design model (9)–(15) featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry (Huber et al., 2018; De Meyer et al., 2015, 2016). Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

References

- Mahsa Arabi, Saeed Yaghoubi, and Javad Tajik. A mathematical model for microalgae-based biobutanol supply chain network design under harvesting and drying uncertainties. *Energy*, 179:1004–1016, 2019. ISSN 0360-5442. doi: 10.1016/j.energy.2019.04.219.
- Marcos Barrio-Anta, Hortensia Sixto-Blanco, Isabel Cañellas-Rey de Viñas, and Fernando Castedo-Dorado. Dynamic growth model for short-rotation poplar plantations in northern and central-western Spain. *Forest Ecology and Management*, 255(3–4):1257–1268, 2008. ISSN 0378-1127. doi: 10.1016/j.foreco.2007.10.026.
- Annelies De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. A generic mathematical model to optimise strategic and tactical decisions in biomass-based supply chains (optimass). *European Journal of Operational Research*, 245(1):247–264, 2015. ISSN 0377-2217. doi: 10.1016/j.ejor.2015.02.042. URL <https://www.sciencedirect.com/science/article/pii/S0377221715001654>.
- Annelies De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. Considering biomass growth and regeneration in the optimisation of biomass supply chains. *Renewable Energy*, 87:990–1002, 2016. ISSN 0960-1481. doi: 10.1016/j.renene.2015.09.038. URL <https://www.sciencedirect.com/science/article/pii/S0960148115301397>.
- Sandra D. Eksioglu, Ambarish Acharya, Liam E. Leightley, and Sumesh Arora. Analyzing the design and management of biomass-to-biorefinery supply chain. *Computers & Industrial Engineering*, 57(4):1342–1352, 2009. ISSN 0360-8352. doi: 10.1016/j.cie.2009.07.003.
- Diego Espinoza, Farid Chejne, José M. Ponce-Ortega, and Francisco Nápoles-Rivera. Sustainable biorefinery supply chain design and management: A review. *Bioresource Technology*, 243:1223–1235, 2017. ISSN 0960-8524. doi: 10.1016/j.biortech.2017.07.089.

Robin J. Faasch and Genevieve Patenaude. The economics of short rotation coppice in Germany. *Biomass and Bioenergy*, 45:27–40, 2012. ISSN 0961-9534. doi: 10.1016/j.biombioe.2012.05.014.

A. R. Graves, P. J. Burgess, J. H. N. Palma, F. Herzog, G. Moreno, M. Bertomeu, C. Dupraz, F. Liagre, K. Keesman, W. van der Werf, A. Koeffeman de Nooy, and J. P. van den Briel. Development and application of bio-economic modelling to compare silvoarable, arable, and forestry systems in three European countries. *Ecological Engineering*, 29(4):434–449, 2007. ISSN 0925-8574. doi: 10.1016/j.ecoleng.2006.09.018.

Henning Grünewald, Reinhold Böcker, B. Ulrich Schneider, Oliver Bens, and Reinhard F. Hüttel. Biomass production of black locust (*Robinia pseudoacacia* L.) short rotation plantations in Germany: growth and economics of different clones and management practices. *Bioresource Technology*, 98(2):281–286, 2007. ISSN 0960-8524. doi: 10.1016/j.biortech.2006.01.004.

Julia A. Huber, Katharina May, and Kurt-Jürgen Hülsbergen. Allometric tree biomass models of various species grown in short-rotation agroforestry systems. *European Journal of Forest Research*, 135(2): 283–296, 2016. ISSN 1612-4669. doi: 10.1007/s10342-015-0932-5.

Julia A. Huber, Michael Matiu, and Kurt-Jürgen Hülsbergen. First-rotation growth and stand structure dynamics of tree species in organic and conventional short-rotation agroforestry systems. *Heliyon*, 4(7): e00645, 2018. ISSN 2405-8440. doi: 10.1016/j.heliyon.2018.e00645. URL <https://www.sciencedirect.com/science/article/pii/S2405844017334904>.

Patrick Lamers, Eric Tan, Erin Searcy, Christopher Scarlata, Kara Cafferty, and Jacob Jacobson. Strategic supply system design – a holistic evaluation of operational and production cost for a biorefinery supply chain. *Biofuels, Bioproducts and Biorefining*, 9(6):648–660, 2015.

P. K. Ramachandran Nair. *An Introduction to Agroforestry*. Kluwer Academic Publishers, Dordrecht, 1993. ISBN 978-0792321347.

Marzena Niemczyk, Mariusz J. Stolarski, Marta Krzyżaniak, et al. The effects of cultivar and rotation length (5 vs. 10 years) on biomass production and sustainability of poplar (*populus* spp.) bioenergy plantation. *GCB Bioenergy*, 13:999–1014, 2021. doi: 10.1111/gcbb.12827.

B. Sharma, R. G. Ingalls, C. L. Jones, and A. Khanchi. A review of biomass supply chain management studies in the context of bioenergy production. *Biomass and Bioenergy*, 56:147–158, 2013. ISSN 0961-9534. doi: 10.1016/j.biombioe.2013.04.005.

Raffaele Spinelli, Carla Nati, Lorenzo Sozzi, Natascia Magagnotti, and Gianni Picchi. Physical characterization of commercial woodchips on the italian energy market. *Fuel*, 90(6):2198–2202, 2011. doi: 10.1016/j.fuel.2011.02.028.

Riccardo Testa, Anna Maria Di Trapani, Mario Foderà, Filippo Sgroi, and Salvatore Tudisca. Economic evaluation of introduction of poplar as biomass crop in Italy. *Renewable and Sustainable Energy Reviews*, 38:775–780, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.07.054. URL <https://www.sciencedirect.com/science/article/pii/S1364032114005473>.

Eric Toensmeier. Perennial staple crops and agroforestry for climate change mitigation. In Florencia Montagnini, editor, *Integrating Landscapes: Agroforestry for Biodiversity Conservation and Food Sovereignty*, volume 12 of *Advances in Agroforestry*, pages 431–453. Springer, Cham, 2017. doi:

- 412 10.1007/978-3-319-69371-2_18.
- 413 Miroslav Trnka, J. Fialová, V. Koutecký, M. Fajman, Z. Šalud, and S. Hejduk. Biomass production and
414 survival rates of selected poplar clones grown under a short-rotation system on arable land. *Plant,*
415 *Soil and Environment*, 54(2):78–88, 2008. ISSN 1214-1178. URL [https://www.agriculturejournals.](https://www.agriculturejournals.cz/web/pse.htm?type=article&id=419_2007-PSE)
416 [cz/web/pse.htm?type=article&id=419_2007-PSE](https://www.agriculturejournals.cz/web/pse.htm?type=article&id=419_2007-PSE).
- 417 Seyed Mohammad Zahraee, Bijan Bahrami, Mehdi Baghery, and Arash Ahmadi. A comprehensive review
418 of biomass supply chain modelling and optimisation. *Renewable and Sustainable Energy Reviews*, 110:
419 1–19, 2019. ISSN 1364-0321. doi: 10.1016/j.rser.2019.04.071.